

TANVI AGARWAL

Amherst, MA | +1 (413) 437-6272 | tanviagarwal@umass.edu | github.com/tanvi217 | linkedin.com/in/tanvi217

EDUCATION

University of Massachusetts, Amherst

Master of Science in Computer Science

Related Coursework: Database Design & Implementation, Machine Learning, Data Science Fundamentals

Massachusetts, USA

January 2025 - December 2026

Indian Institute of Information Technology, Sri City

BTech in Electronics & Communication Engineering: CGPA: 8.87/10

Chittoor, India

July 2017 - June 2021

WORK EXPERIENCE

Microsoft

Hyderabad, India

Software Engineer II - OneDrive & SharePoint Platform Team | C#, C++, .NET, SQL

September 2023 - December 2024

- Co-designed and developed the end-to-end [Restricted Access Control](#) policy in C++ SharePoint codebase with a 2-member team, collaborating across teams—preventing oversharing for 3.5M users and generating \$4M in revenue at launch.
- Implemented core group-based access control and PowerShell cmdlets for policy enablement, reducing evaluation time by 40% through optimized SQL stored procedures, distributed caching and query indexing.
- Architected a sharing blocker, that cut unauthorized access denials by 76%, leveraging batch-processed external API calls, in-memory caching and a throttling mechanism.
- Developed policy usage analytics, processing 10M+ records daily, using optimized SQL queries, Scala and Apache Spark on Azure Data Lakes for ETL workflows.
- Migrated the *Just-In-Time Data Loss Prevention* microservice to the latest Azure Functions, reducing infra costs by 25% and improving reliability through Kubernetes auto-scaling.

Software Engineer - Office Web Experiences | C#, .NET, Kusto, Azure services

July 2021 - August 2023

- Reduced API latency by 40% by onboarding *Reply @Mentions* microservice to MSAL for secure S2S authentication and using distributed Redis caching. Built a Kusto-based dashboard to monitor reliability and latency metrics.
- Improved Planner-Excel task synchronization by designing a decoupled workflow using session-based Azure Service Bus and automating resource provisioning with ARM templates, reducing configuration errors.
- Cut manual effort by 75% by automating periodic monitoring of the workflow to ensure in-order task synchronization.

Amazon

Hyderabad, India

Software Development Engineer Intern - Fast Data Technologies | Java, Spring Boot, AWS

January 2021 - June 2021

- Reduced query execution time by 35% by implementing paginated RESTful APIs, refining query logic, and leveraging DynamoDB's GSI for efficient access, while writing integration tests for robust functionality.
- Optimized throughput by creating a cron job to monitor and detect stalled pipelines utilizing multi-threading.

PROJECTS

- IceBreaker:** Led a team of four to develop a Django-based crowdfunding platform with role-based authentication, real-time notifications, and Stripe payment gateway integration. Designed a scalable MySQL database, built RESTful APIs, and integrated the backend with the frontend.
- SpeakerOre:** Built using TypeScript and ReactJS to provide event leads to subscribed users; launched on AWS EC2 with responsive design and optimized performance. Used React hooks for state management and app efficiency improvement.

LEADERSHIP & ACHIEVEMENTS

- Microsoft Fix-Hack-Learn(FHL) 2022:** Ranked top two for the *Data Boundary Compliant ML Models Training* project.
- Adobe Codiva Hackathon 2020:** Ranked among the top 100 coders.
- Women Techmakers Engineering Fellow 2019 (Program supported by Google):** Selected among 100 women nationwide for a month-long boot camp with Google leaders and awarded a scholarship.
- Mentorship:** Led workshops on Applied Software Engineering as a member of IIIT Sricity's IOTA club; mentored a Microsoft summer intern on *Intelligent Replies with @Mentions* project for Office apps.
- Teaching Assistant:** Facilitated the *Signals & Systems* course guiding freshers in MATLAB programming assignments.

TECHNICAL SKILLS

Languages & Databases: C, C++, C#, Java, JavaScript, Python, R, HTML, CSS, MySQL, PostgreSQL, MongoDB

Frameworks & Libraries: .NET, Spring Boot, ReactJS, Node, Django, PyTorch, Pandas, Azure, Kubernetes

Developer Tools: Git, CI/CD, VS Code, Visual Studio, PyCharm, IntelliJ, Docker